

3.7 Nitrogen Compounds

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.7 Nitrogen Compounds
Difficulty	Hard

Time allowed: 10

Score: /5

Percentage: /100

Question 1

What is the correct two-step synthesis to produce 2-hydroxybutanenitrile from propan-1-ol?

- A. 1) NaCl / H₂SO₄
2) NaCN / Ethanol
- B. 1) NaCN / Ethanol
2) NaCl / H₂SO₄
- C. 1) NaCN (aq) / H⁺ (aq)
2) K₂Cr₂O₇ / H₂SO₄ under distillation
- D. 1) K₂Cr₂O₇ / H₂SO₄ under distillation;
2) NaCN (aq) / H⁺ (aq)

[1 mark]

Question 2

Butanone can undergo addition when mixed with NaCN (aq) / H⁺ (aq), forming product **Z**. Product **Z** can then undergo hydrolysis when heated with dilute HCl, forming product **Y**.

Which compound is product Y?

- A. 2-hydroxypentanoic acid
- B. 2-hydroxybutylamine
- C. 2-methyl-2-hydroxybutanoic acid
- D. 2-ethyl-2-hydroxypropylamine

[1 mark]

Question 3

What is the correct set of reagents and conditions for the two reactions shown?

	(CH ₃) ₂ CO → (CH ₃) ₂ CHCN	CH ₃ CHC/CH ₃ → CH ₃ CH(NH ₂)CH ₃
A	Sn / conc. H ₂ SO ₄	Dilute ethanolic NH ₃
B	Sn / conc. HCl	Excess ethanolic NH ₃
C	NH ₃	Nitric acid / 50 °C
D	KCN, H ₂ SO ₄	Excess ethanolic NH ₃

[1 mark]

Question 4

A student wanted to produce a hydroxynitrile. Which types of reactions, reactants and reagents are shown in the table would correctly describe this reaction?

	Type of reaction	Reactants	Reagents
A	Addition	Aldehyde	HCN
B	Hydrolysis	Halogenoalkane	Ethanolic NH_3
C	Substitution	Ketone	HCN
D	Addition	Halogenoalkane	KCN, ethanol

[1 mark]

Question 5

What would **X**, **Y** and **Z** be in the following reaction sequence?



	X	Y	Z
A	$\text{CH}_3\text{CH}_2\text{CH}_2\text{K}$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{COONa}$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
B	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CN}$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{COONa}$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
C	$\text{CH}_3\text{CH}_2\text{CH}_2\text{K}$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{COONa}$
D	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CN}$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{COONa}$

[1 mark]

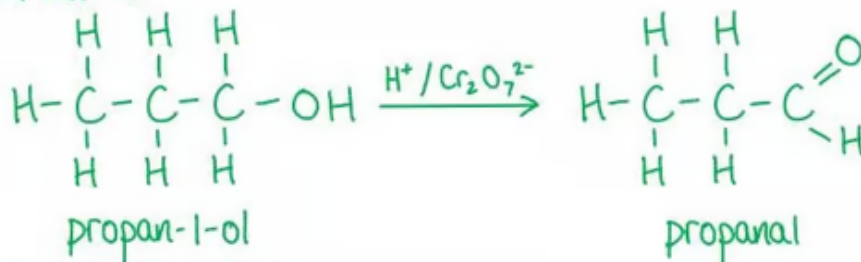
Model Answers: Hard

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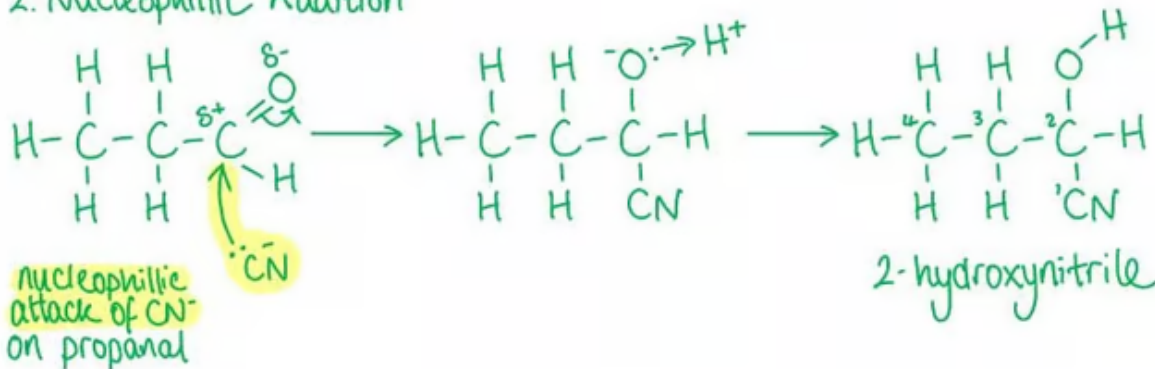
The correct answer is **D** because:

- 2-hydroxybutanenitrile is a **hydroxynitrile**.
- **Hydroxynitriles** can be formed from the **nucleophilic addition** reaction of **aldehydes** and **ketones** with **HCN**, **KCN** as catalyst and heat.
- A **nucleophilic addition** is when a **nucleophile** is added to an unsaturated bond to form a **single** product – a nucleophile donates a pair of electrons.
- Since propan-1-ol is a **primary alcohol** it should be first oxidised to an aldehyde using **acidified potassium dichromate under distillation** to form **propanal**.
- The aldehyde **propanal** is then reacted with **hydrogen cyanide** (acidified NaCN) and heated to produce **2-hydroxybutanenitrile**.

1. Oxidation



2. Nucleophilic Addition



A is incorrect as the reaction of propan-1-ol with $\text{NaCl}/\text{H}_2\text{SO}_4$ will result in a **nucleophilic substitution** reaction resulting in the removal of the OH group so the

compound cannot be **oxidised** to an aldehyde.

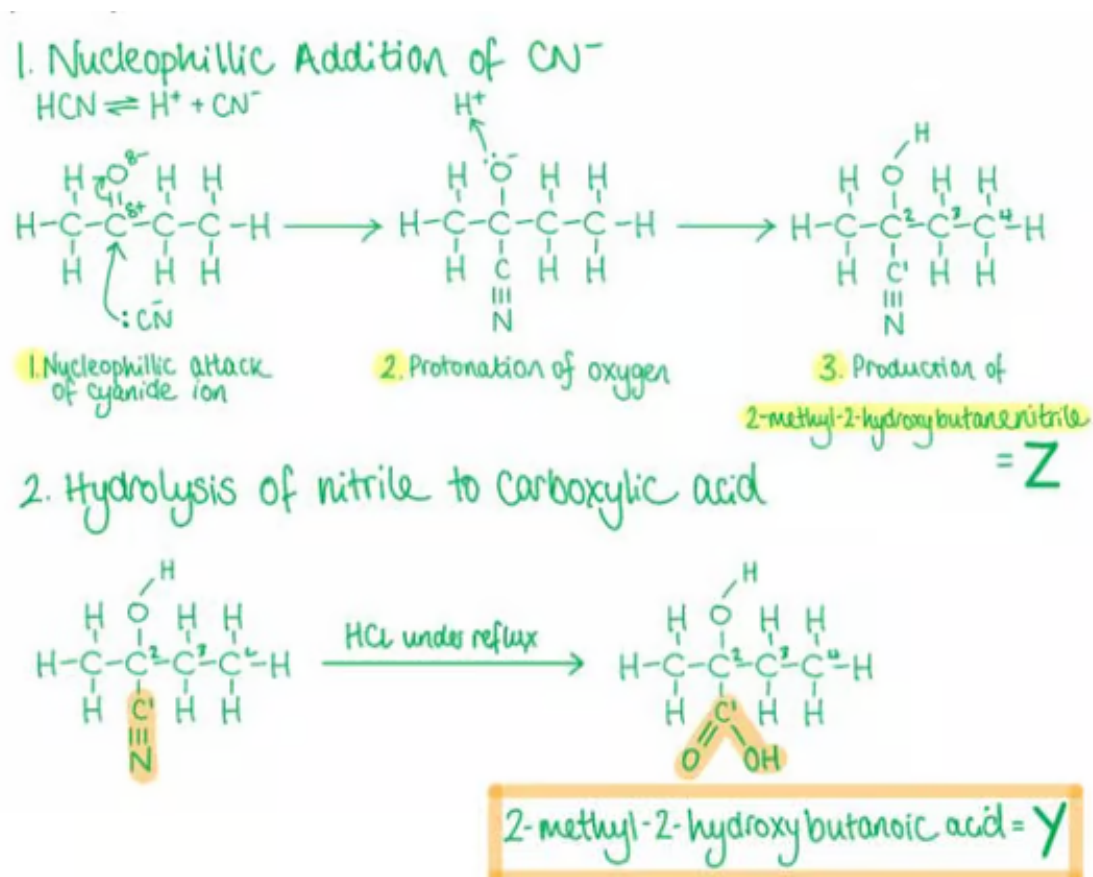
B is incorrect as this is similar to reaction **A** but the steps are in the wrong order. The OH group is replaced so **oxidation** to an aldehyde cannot take place.

C is incorrect as the steps are in the **wrong order** – the CN^- group will only attack a **carbonyl carbon atom** so propan-1-ol need to be oxidised first.

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The correct answer is **C** because:

- A **nucleophilic addition** is when a **nucleophile** is added to an unsaturated bond to form a **single** product.
- **Hydroxynitriles** are formed from the nucleophilic addition reaction of **aldehydes** and **ketones** with **HCN**, **KCN** as catalyst and heat.
- **Butanone** is a **ketone** and will form **2-ethyl-2-hydroxypropanenitrile**.
- **Hydrolysis** is the breakdown of a compound using water.
- **Nitriles** can be hydrolysed by **refluxing** it with **dilute hydrochloric acid**.
- The $-\text{CN}$ group is converted into a $-\text{COOH}$ group, forming a **carboxylic acid**.
- The ammonia in solution will act as a **base** and form an **ammonium salt** with $\text{C}/\text{NH}_4\text{C}/$.



A & B are incorrect as this is the product if the reactant is an **aldehyde**.

D & B are incorrect as this is the product if the CN^- group in Z is **reduced**.

3

The correct answer is **D** because:

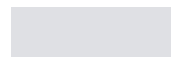
- This is a **nucleophilic addition reaction** in which a **nucleophile** is added to an unsaturated bond to form a **single** product.
 - A nucleophile is a species that donates a pair of electrons.
- In the first reaction, the ketone **propanone** undergoes a **nucleophilic addition** reaction (CN^- is the nucleophile) with acidified KCN to form **isobutylnitrile**.
- In the second reaction, a **secondary halogenoalkane** changes into a **secondary amine**.
- This is a **nucleophilic substitution reaction** in which a nucleophile replaces another nucleophile in the compound.

- In this reaction, **ammonia** acts as the **nucleophile** and replaces the **chlorine** in halogenoalkane to form a secondary amine.

A, B & C are incorrect as they do not show the correct set of reagents and conditions for the two reactions above.

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The correct answer is **A** because:



- In this reaction, **ammonia** acts as the **nucleophile** and replaces the **chlorine** in halogenoalkane to form a secondary amine.

A, B & C are incorrect as they do not show the correct set of reagents and conditions for the two reactions above.

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The correct answer is **A** because:

- A **nucleophilic addition** is when a **nucleophile** is added to an unsaturated bond to form a **single** product.
 - A nucleophile is a species that donates a pair of electrons.
- **Hydroxynitriles** are formed from the **nucleophilic addition** reaction of **aldehydes** and **ketones** with **HCN, KCN** as catalyst and heat.
- Option A shows the reaction type, reactants and reagents correctly.

B is incorrect as this is a **nucleophilic substitution** reaction to form an **amine**.

C is incorrect as this is an **addition reaction** to form an **hydroxynitrile**.

D is incorrect as this is a **nucleophilic substitution** reaction to form a **nitrile**.

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The correct answer is **B** because:

- The reaction of a **halogenoalkane** with ethanolic KCN and heat is a **nucleophilic substitution** reaction to form a nitrile.
 - A nucleophilic substitution reaction is a reaction in which a nucleophile replaces another nucleophile.
 - A nucleophile is a species that can donate a pair of electrons.
- Therefore, only option **B** and **D** can be correct.
- The reaction of nitriles with aqueous alkali solution when heated is a **hydrolysis reaction** to produce a **sodium salt (sodium ethanoate)** and **ammonia**.
 - In a **hydrolysis** reaction a compound is broken down by **water**.

- The sodium ethanoate is acidified with **dilute hydrochloric acid**.
- The **ethanoate ion** in the **sodium ethanoate salt** will react with the hydrogen ions to form **ethanoic acid**.
 - Once reacted, ethanoic acid is a weak acid and holds onto the hydrogen rather than dissociating.
- Therefore, only option **B** is correct.

A & C are incorrect as compound **X** is not a **nitrile** and option **C** has compounds **Y** and **Z**